

Crypto Quant Strategy — Research Report

A 3-Component Momentum/Reversal Ensemble with a Correlation-Regime Risk Overlay

Executive Summary

This report presents an all-weather crypto trading strategy on 53 liquid Binance USDT pairs. The headline strategy is a 3-signal ensemble — Volume Momentum, Smart TSMOM, and a Vol-Adjusted Intraday Reversal — combined by inverse-variance weighting and sized by a correlation-regime overlay.

On the out-of-sample test (2023-07 → 2025-12), net of realistic costs, it delivers a **Sharpe ratio of 1.56**, a maximum drawdown of **-10.7%**, and a Calmar of **1.55** — versus BTC Buy & Hold's 1.15 Sharpe at roughly 4× the volatility and 3× the drawdown. Because Sharpe is leverage-invariant, the same edge scales from the conservative ~11%-volatility presentation up to ~31%/yr at a 20% volatility target with no borrowing (Section 7.3).

Strategy	SR train	SR test	Max DD test	Calmar test
BTC Buy & Hold	0.37	1.15	-32.0%	1.64
BTC + vol-targeting	0.39	0.86	-18.5%	1.01
TSMOM Vol-Managed (baseline)	1.62	0.72	-24.2%	0.71
Smart TSMOM VM	0.84	0.92	-36.8%	0.79
Volume Momentum Binary VM	1.20	0.76	-29.3%	0.57
Vol-Adjusted Reversal	0.37	0.81	-32.3%	1.28
2c Ensemble + Overlay	1.59	1.37	-11.7%	1.42
★ 3c Ensemble + Overlay ★	1.74	1.56	-10.7%	1.55

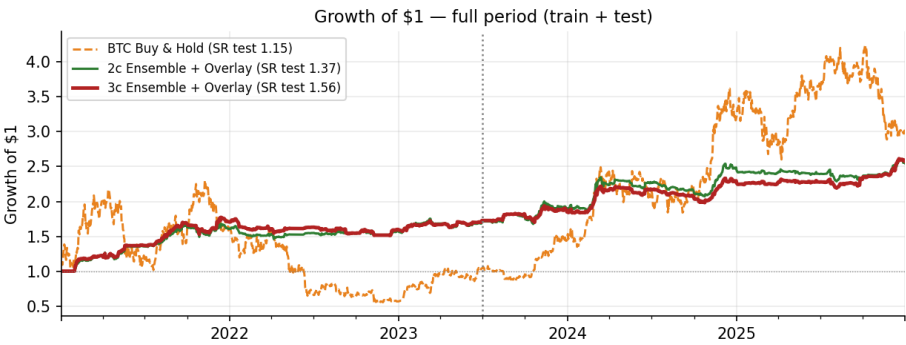


Figure 1. Growth of \$1 over the full period — the 3c ensemble (dark red) vs the 2c ensemble and BTC. The ensemble grows more slowly in absolute terms because it runs at far lower volatility; its advantage is risk-adjusted, and the level is a leverage choice (Section 7.3).

1. Objective & Scope

The goal is a cryptocurrency strategy that holds up out-of-sample across a full market cycle, built on signals with a clear economic rationale and evaluated net of realistic trading costs. The report develops, for each component, **what it does** and **why it works** in theory, then **why combining them** improves the risk-adjusted return. The strategy is implemented in `strategy.ipynb`.

2. Data & Universe

Source. Daily and 6-hour OHLCV from Binance spot, 2021-01-01 → 2025-12-31.

Universe. From ~64 USDT pairs that existed at the start of 2021, two filters apply: (1) full price history across train + test, and (2) median daily quote volume > \$5M during train. This yields **53 coins** — a cross-section with meaningful dispersion while avoiding micro-cap slippage. Crypto factor studies use comparably liquid universes (Cong, Karolyi, Tang & Zhao 2023).

Train/test split. Train 2021-01 → 2023-06 (the 2021 bull, the 2022 bear, the March-2023 banking stress); test 2023-07 → 2025-12 (recovery, the spot-ETF rally, the 2024–25 ATH cycle). The two windows span structurally different regimes — the central test of robustness.

Return characteristics (train). Daily log-returns are strongly non-normal (pooled excess kurtosis ≈ 9.0 , skew ≈ 0.28), so Calmar and max-drawdown are reported alongside Sharpe. Daily return autocorrelation is weak (Figure 3a) — momentum lives in N -day *cumulative* trends, not one-day persistence — while the autocorrelation of squared returns is strongly positive (Figure 3b): **volatility clusters**, which is what makes volatility forecastable and vol-scaling effective.

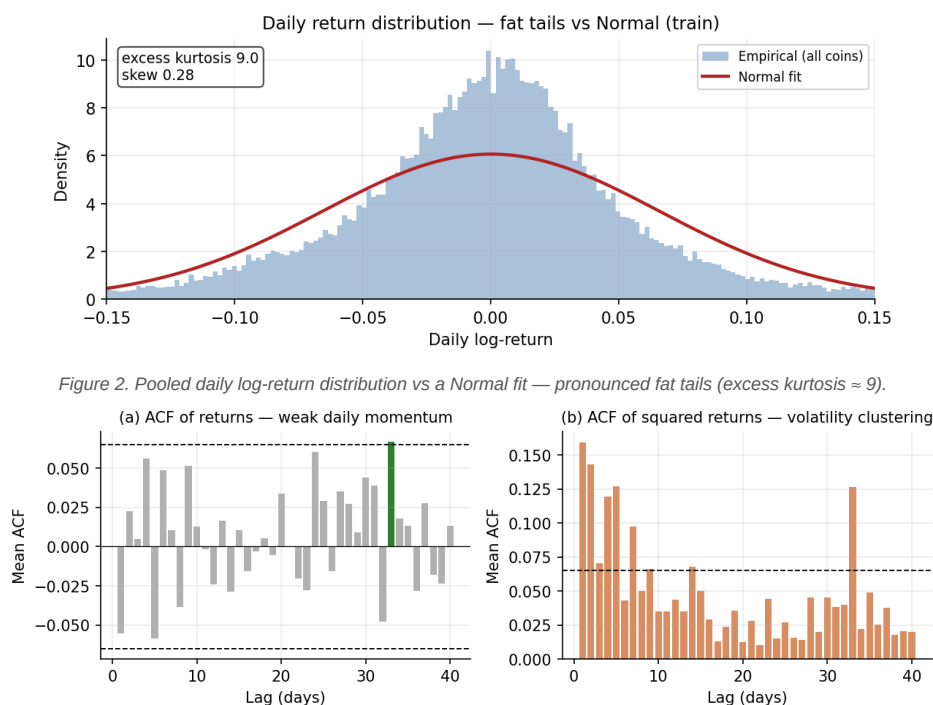


Figure 3. Daily autocorrelation: (a) returns — weak momentum (~1 of 40 lags significant); (b) squared returns — strong volatility clustering.

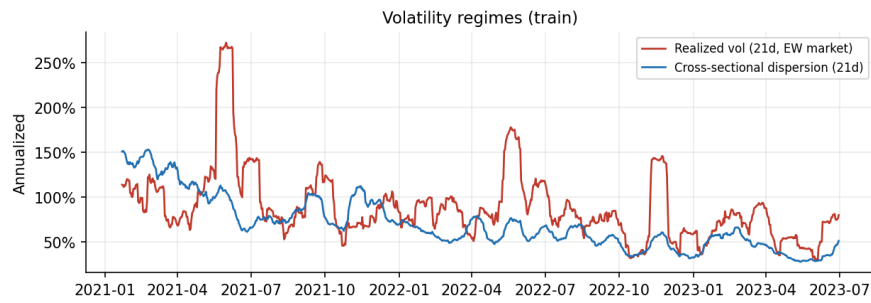


Figure 4. Realized volatility and cross-sectional dispersion over the train window — both vary substantially, motivating volatility scaling.

Market regimes. The training window is predominantly bearish: BTC spent ~68% of it below its 200-day moving average and ~80% in a drawdown deeper than 20% (the 2022 bear). Fitting on a mostly bearish window and testing on the 2023–25 recovery is a demanding regime contrast — and the reason the strategy favours signals whose causal direction is stable across regimes (Section 4).

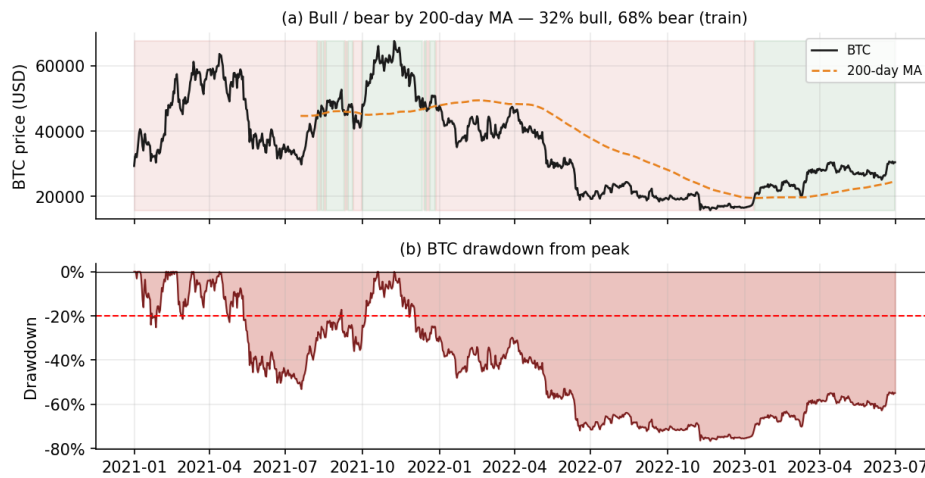


Figure 5. Market regimes in train: (a) BTC vs its 200-day MA with bull (green) / bear (red) shading; (b) drawdown from peak.

Universe correlation. The 53 coins are highly co-moving — mean pairwise return correlation is 0.56 (range [0.33, 0.85]) over the train window, and it rises further during stress. This is the condition the correlation-regime overlay exploits (Section 5.3): when everything moves together, the cross-sectional edge collapses and exposure should be cut.

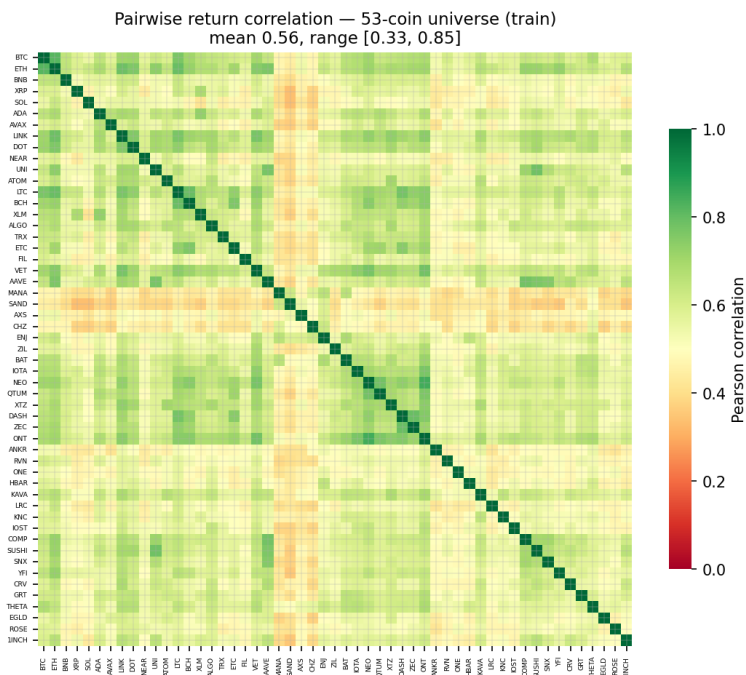


Figure 6. Pairwise return-correlation matrix of the 53-coin universe (train) — uniformly positive, mean 0.56.

3. Backtesting Framework & Cost Model

Positions are normalized to unit gross leverage and applied with a one-day lag (`w.shift(1)`): the weight formed from information available on day t is applied to the day $t+1$ return.

All results are net of **15 bps/side** for daily strategies (10 bps Binance taker + 5 bps slippage) and **20 bps/side** for the intraday reversal (wider spreads during stress events). Costs are material — they cost the ensemble roughly 0.8 of Sharpe gross-to-net, with break-even near 45 bps/side (Section 7.2).

4. Signal Construction

Each signal is a per-asset, market-neutral-leaning timing rule. We describe what it does, then the theory of why it earns a return.

The per-signal charts below plot raw growth of $\backslash\$1$ against BTC. Because every signal runs at a different (and mostly lower) volatility than BTC, the curves are **not** at equal risk — the dollar levels are a leverage artefact, so the fair comparison is the **Sharpe** in each legend, not the ending balance. (The headline strategy's volatility and leverage choices are treated in Section 7.3.)

4.1 Time-Series Momentum (baseline)

Reference: Moskowitz, Ooi & Pedersen (2012), "Time Series Momentum", JFE.

What it does. Each coin is traded on the sign of its own 30-day cumulative return — long after a positive run, short after a negative one — and the position is scaled by the coin's realized volatility so every name contributes roughly equal risk.

Construction —

$$S_{i,t} = \sum_{k=0}^{N-1} r_{i,t-k}, \quad N=30$$

$$\text{sig}_{i,t} = \text{sign}(S_{i,t}) 1\{|S_{i,t}| > \theta\}, \quad \theta = 0.02$$

$$w_{i,t} = \frac{\text{sig}_{i,t}}{\sigma_{i,t}}, \quad \sigma_{i,t} = \text{std}(r_i, 20 \text{ d})$$

Why it works. Time-series momentum is one of the most pervasive anomalies in finance, documented across decades and asset classes (MOP 2012). The accepted mechanisms are behavioural and flow-based: (i) **under-reaction** — information diffuses gradually and investors update beliefs slowly, so prices trend toward fair value rather than jumping to it; (ii) **positive-feedback trading** — trend-followers, CTAs and, in crypto, momentum-chasing retail flows extend moves once they start. Both forces are amplified in crypto by retail dominance and slow, fragmented information processing, making trend persistence strong at the monthly horizon (N=30).

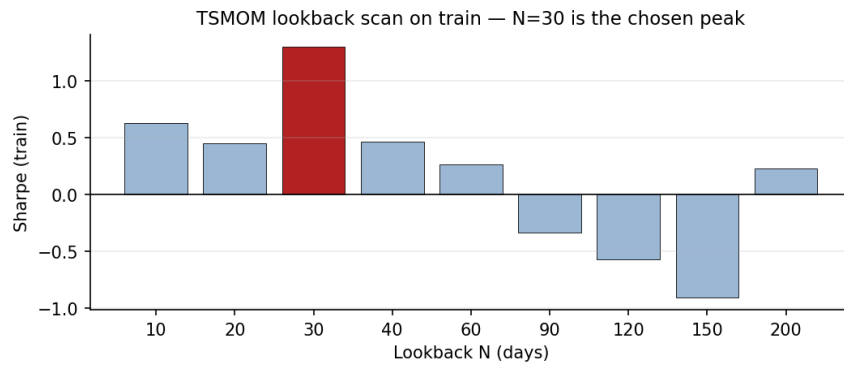


Figure 7. Train Sharpe by lookback N — the monthly horizon (N=30) is the strongest; very short and very long lookbacks add noise or stale signal.

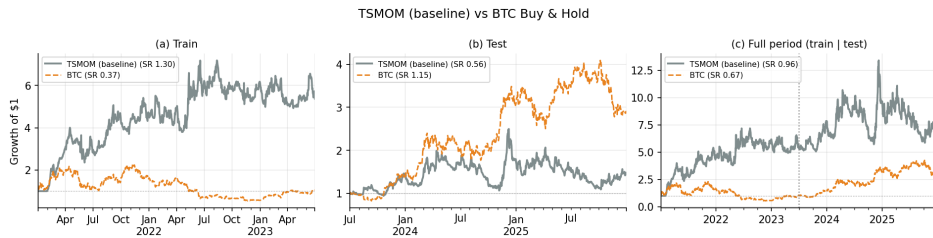


Figure 8. TSMOM (baseline) vs BTC Buy & Hold — (a) train, (b) test, (c) full period compounded continuously (dotted line = train/test split). Each panel's legend shows the Sharpe over that window. This raw baseline is unscaled and runs at ~71% annualized volatility, so its dollar-growth dominance reflects leverage, not skill — the fair read is the risk-adjusted Sharpe. On that basis it beats BTC on train (1.30 vs 0.37) but decays out-of-sample (0.56 vs 1.15), motivating the vol-scaling and filters that follow.

4.2 Volatility-Managed TSMOM

References: Moreira & Muir (2017), "Volatility-Managed Portfolios", JF; Hurst, Ooi & Pedersen (2017) for the linear vol-targeting variant used here (constant-vol scaling rather than MM's $1/\sigma^2$ formulation); Grobys (2025) on crypto momentum crashes.

What it does. The whole momentum book is scaled up or down so that its realized volatility tracks a constant target (~19% annualized); the scaling uses a lagged forecast of the strategy's own volatility, so high-vol periods get less capital and calm periods get more.

Construction —

$$\hat{\sigma}_t = \text{std}(R_{t-20:t-1}) \quad (\text{lagged})$$

$$L_t = \text{clip}\left(\frac{\sigma^*}{\hat{\sigma}_t}, 0, 3\right), \quad \sigma^* = 0.01/\text{day}$$

$$R_t^{\text{VM}} = L_t R_t$$

Why it works. Volatility is highly persistent and forecastable (the clustering in Figure 3b), **but expected returns do not rise one-for-one with volatility**. Selling exposure when forecast volatility is high — and buying it when low —

therefore improves the realized Sharpe (Moreira-Muir). The mechanism matters most for momentum specifically: "momentum crashes" cluster in high-volatility panic-rebound regimes (the sharp bear-market snapbacks), and vol-scaling cuts exposure exactly there, removing the fattest left tail of the raw strategy (Grobys, in crypto). Result: the net-of-cost Sharpe rises from a raw 1.30 to 1.62 on train (the raw baseline runs at ~71% annualized volatility; vol-scaling also cuts that to ~19%) and the worst drawdown shrinks materially.

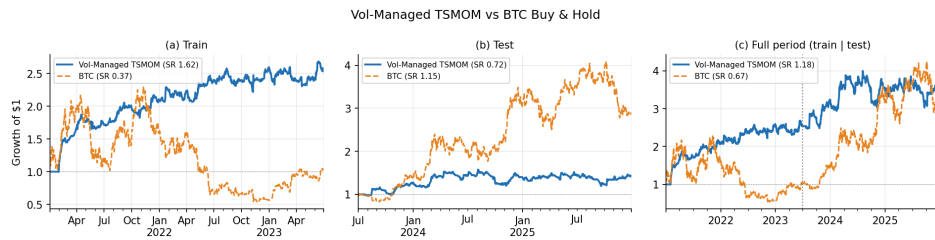


Figure 9. Vol-Managed TSMOM vs BTC — train, test, and full period. Vol-scaling lifts the train Sharpe to 1.62 and smooths the equity curve relative to the raw baseline (Figure 8).

4.3 Smart TSMOM

Reference: Nguyen (2026), "AdaptiveTrend" (arXiv:2602.11708), which reports SR 2.41 with a 6h trend-follower, monthly re-optimization, trailing stops and 150+ pairs.

What it does. Two refinements on the vol-managed baseline: (i) a **rolling-Sharpe filter** — a coin is traded only if its own 30-day rolling Sharpe exceeds ± 0.3 ; and (ii) an **asymmetric 70/30** long-short tilt (longs $\times 1.4$, shorts $\times 0.6$).

Construction —

$$\text{Sh}_{i,t} = \frac{\text{mean}(r_i, 30 \text{ d})}{\text{std}(r_i, 30 \text{ d})}$$

$$\text{sig}_{i,t}^{\text{filt}} = \text{sig}_{i,t}^{\text{TSMOM}} \cdot 1\{\text{sig}_{i,t}^{\text{TSMOM}} \cdot \text{Sh}_{i,t} > 0.3\}$$

$$w_{i,t} = \frac{\text{sig}_{i,t}^{\text{filt}}}{\sigma_{i,t}} \cdot m, \quad m = 1.4 (\text{long}), 0.6 (\text{short})$$

Why it works. (i) Momentum pays when a trend is persistent and smooth and loses to whipsaw when it is choppy. The rolling-Sharpe filter is a *trend-quality* conditioner: it keeps positions only in coins whose recent move is large relative to its noise, discarding the low-information chop where momentum reverses. (ii) Crypto carries a structural positive drift (adoption and a large risk premium), so a dollar-neutral long-short discards that upward bias; tilting net-long captures the drift while shorts still hedge cross-sectional dispersion. Standalone SR test rises to 0.92 (from 0.72), though its edge is mostly the long-side drift (Section 6).

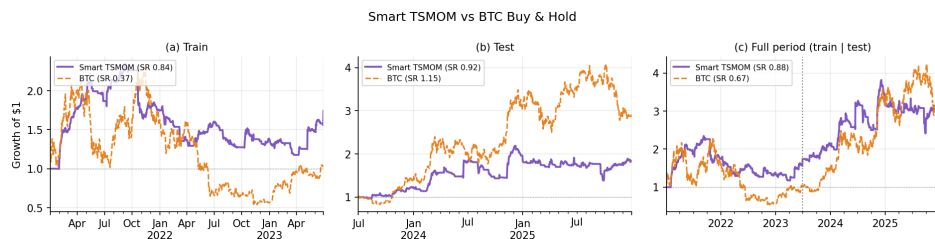


Figure 10. Smart TSMOM vs BTC — train, test, and full period. Unlike the baselines, its risk-adjusted edge is stable across the split (SR train 0.84, test 0.92).

4.4 Volume Momentum Binary — the primary alpha

References: Gervais, Kaniel & Mingelgrin (2001), "The High-Volume Return Premium", JF — assets experiencing unusually high own-volume tend to appreciate over the following weeks; Lee & Swaminathan (2000), "Price Momentum and Trading Volume", JF — high-volume past winners continue more strongly while low-volume winners reverse, establishing the volume $\times$ price-momentum interaction this signal exploits.

What it does. For each coin we compute the 30-day change in log-volume, z-score it against the coin's own 60-day history, and go **long when own-volume is rising ($z > 0.5$)**, **short when collapsing ($z < -0.5$)** — a binary, volatility-scaled position.

Construction —

$$\Delta V_{i,t} = \log V_{i,t} - \log V_{i,t-30}$$

$$z_{i,t} = \frac{\Delta V_{i,t} - \text{mean}_{60}(\Delta V_i)}{\text{std}_{60}(\Delta V_i)}$$

$$\text{sig}_{i,t} = \text{sign}(z_{i,t}) \mathbb{1}\{|z_{i,t}| > 0.5\}, \quad w_{i,t} = \frac{\text{sig}_{i,t}}{\sigma_{i,t}}$$

Why it works. Volume is information about the *conviction* behind a price move. The volume–return literature (the high-volume return premium; volume confirming trend) shows that moves accompanied by rising participation reflect genuine information and order flow and tend to *continue*, while low-volume drifts fade. The key property is that this is a **per-coin, time-series** relation: "a coin's own volume is rising $\rightarrow$ its price tends to continue" has a **causally stable direction in both bull and bear markets**. That is the decisive difference from a cross-sectional volume rank (which coin has the most volume *relative to others*), whose sign inverts between regimes — the reason a 12-factor cross-sectional "Factor Zoo" we tested collapses out-of-sample (SR test ≈ -1.1). This regime-stability is why Volume Momentum is the most robust signal and the backbone of the ensemble (its train-to-test Sharpe holds across a wide parameter range).

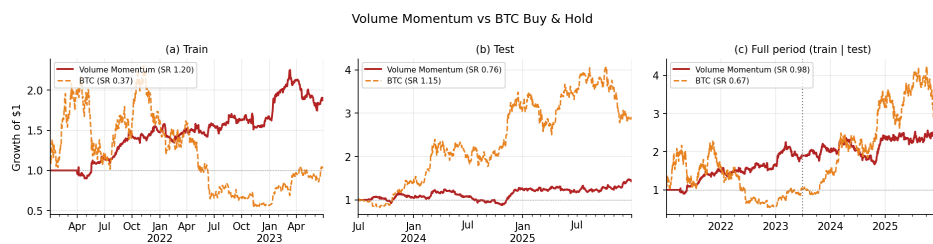


Figure 11. Volume Momentum vs BTC — train, test, and full period. The primary alpha source: it carries the lower-volatility curve but holds a positive risk-adjusted edge in both windows (SR train 1.20, test 0.76), nearly uncorrelated with BTC.

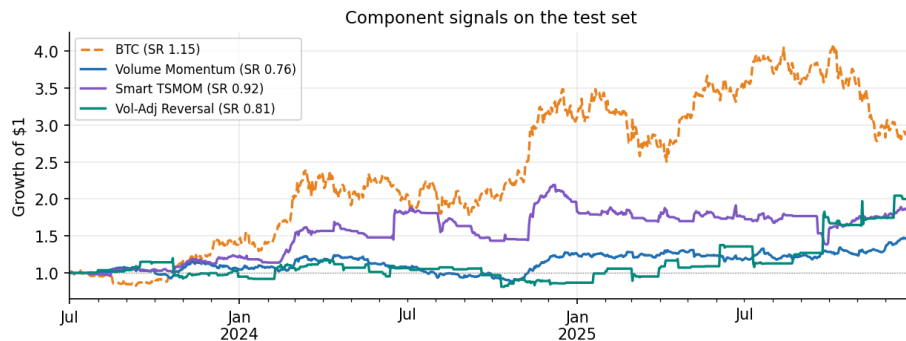


Figure 12. The component signals on the test set (growth of \$1). Volume Momentum and Smart TSMOM are daily; the reversal is daily-aggregated.

4.5 Vol-Adjusted Intraday Reversal

References: Heston, Korajczyk & Sadka (2010), JF, on intraday return patterns driven by liquidity; Jegadeesh (1990) and Lehmann (1990) on short-horizon reversal.

What it does. On 6-hour bars, when a coin's 24-hour cumulative return exceeds ± 4 standard deviations of its own recent volatility, we take the **opposite** position. The threshold is vol-adaptive, so it fires only on genuine tail events ($\sim 1.5\%$ of bar-coin observations overall — 2.6% in train, 0.4% in test, reflecting a calmer test regime).

Construction (6-hour bars) —

$$C_{i,t} = \sum_{k=0}^{\ell-1} r_{i,t-k}, \quad \ell = 4 \text{ bars (24h)}$$

$$\tau_{i,t} = k \sqrt{\ell} \text{ std}(r_i, 120), \quad k = 4$$

$$\text{sig}_{i,t} = -\text{sign}(C_{i,t}) 1\{|C_{i,t}| > \tau_{i,t}\}, \quad w_{i,t} = \frac{\text{sig}_{i,t}}{\text{std}(r_i, 80)}$$

Why it works. Very large short-horizon moves are mostly **liquidity events, not information**: order-flow imbalances, forced liquidations and thin order books push price away from fair value, and as liquidity is replenished the move reverts. Short-term reversal is the classic compensation earned by **liquidity providers** for absorbing these imbalances (Lehmann 1990; Jegadeesh 1990), and Heston-Korajczyk-Sadka (2010) tie intraday return patterns to liquidity dynamics. The 4σ filter isolates the extreme-overreaction tail, where reversal is strongest and least contaminated by real news. Standalone SR test 0.81 (net of 20 bps), and — crucially for the ensemble — **near-zero correlation** with the daily signals (it trades a different mechanism at a different frequency). It is the lowest-confidence component (few triggers) and is capped at 20%.

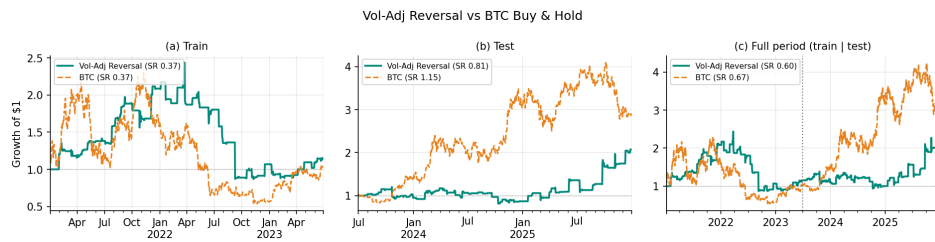


Figure 13. Vol-Adjusted Reversal vs BTC — train, test, and full period (6h signal aggregated to daily). A flat-then-jump profile typical of a rare-event strategy: it sits in cash between triggers and is essentially uncorrelated with BTC (SR train 0.37, test 0.81).

5. The Ensemble — Why Combining Strategies Wins

5.1 The theory of combining strategies

The central reason to run several signals together is **diversification of timing risk**. Consider K strategies, each with the same standalone Sharpe s and average pairwise correlation ρ . An equal-risk combination has Sharpe

$$\text{SR} \approx s \cdot \sqrt{K / (1 + (K-1)\rho)}.$$

Two limits make the intuition clear. If the strategies are perfectly correlated ($\rho = 1$), combining them adds nothing — SR stays s . If they are **uncorrelated** ($\rho \rightarrow 0$), $\text{SR} \approx s \cdot \sqrt{K}$: each extra independent signal raises the Sharpe by the square root of breadth. This is the portfolio analogue of the *fundamental law of active management* — information ratio grows with the number of **independent** bets. The lever that matters is therefore **low correlation between signals, not a high standalone Sharpe of any one of them**.

Our three signals are deliberately built on different mechanisms (price trend, volume conviction, liquidity reversal) at different frequencies, and they are near-orthogonal — pairwise correlation ≤ 0.19 , and ≈ 0 for the reversal. Plugging $K = 3$ and small ρ into the formula predicts a Sharpe roughly $1.5\text{--}1.7\times$ the average component — which is what we observe: standalone Sharpes of $0.76\text{--}0.92$ combine to **1.37** (pre-overlay) and **1.56** (with overlay), well above any single signal.

Inter-signal correlation (test)	Smart TSMOM	Volume Mom.	Reversal
Smart TSMOM	1.00	0.19	-0.05
Volume Mom.	0.19	1.00	-0.01
Reversal	-0.05	-0.01	1.00

5.2 Inverse-variance weighting

Reference: López de Prado, *Hierarchical Risk Parity*.

When signals are near-uncorrelated, the variance-minimizing (and maximum-Sharpe) combination reduces to **inverse-variance weighting**: weight each signal by $1/\sigma^2$ so that all contribute equal risk. We estimate the variances on train only. This is deliberately the simplest robust choice — it needs no covariance matrix to estimate (and therefore nothing to overfit), and it prevents the lowest-variance signal from dominating. The reversal is additionally capped at 20% so the lowest-confidence component cannot take over by virtue of its low variance. Resulting weights: Volume Momentum $\approx 52\%$, Smart TSMOM $\approx 28\%$, Reversal = 20%.

Weights —

$$\omega_k = \frac{1/\sigma_k^2}{\sum_j 1/\sigma_j^2}, \quad \omega_{\text{rev}} \leq 0.20$$

5.3 Correlation Regime Overlay

References: Longin & Solnik (2001) on extreme correlations in crises; Kritzman, Li, Page & Rikkonen (2011), the systemic-risk "Absorption Ratio".

Diversification breaks down precisely when it is most needed: in a crisis, cross-sectional correlation spikes toward 1, the signals stop being independent, and the ensemble degenerates into one big directional bet. The overlay manages this **residual systematic risk**: it scales gross exposure down as the rolling 30-day average pairwise correlation rises (linearly between a low and a high threshold).

Overlay rule —

$$s_t = \text{clip}\left(\frac{c_{\text{hi}} - \bar{\rho}_{t-1}}{c_{\text{hi}} - c_{\text{lo}}}, 0, 1\right), \quad R_t^{\text{ov}} = s_t R_t^{\text{ens}}$$

It is risk sizing, not new alpha — and in train, ensemble Sharpe by correlation tercile is 3.00 (low), 0.14 (mid), 1.08 (high), confirming the edge lives on low-correlation days.

The band (0.55, 0.85) is chosen by a one-standard-error rule on train (the least-aggressive band whose Sharpe is within 1 SE of the best), deploying $\sim 52\%$ of capital on average. It lifts the ensemble from SR test 1.07 (MDD -23%) to **1.37 (MDD -12%)**. In short: inverse-variance weighting diversifies the *idiosyncratic* risk of the signals; the overlay manages the *systematic* risk that diversification cannot.

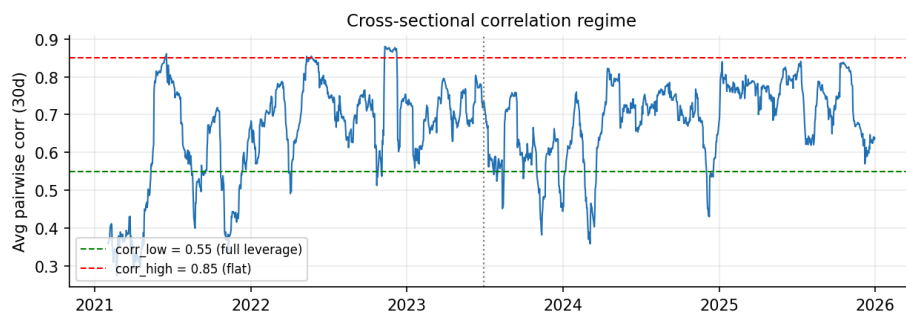


Figure 14. Average pairwise correlation (30-day) with the overlay thresholds.

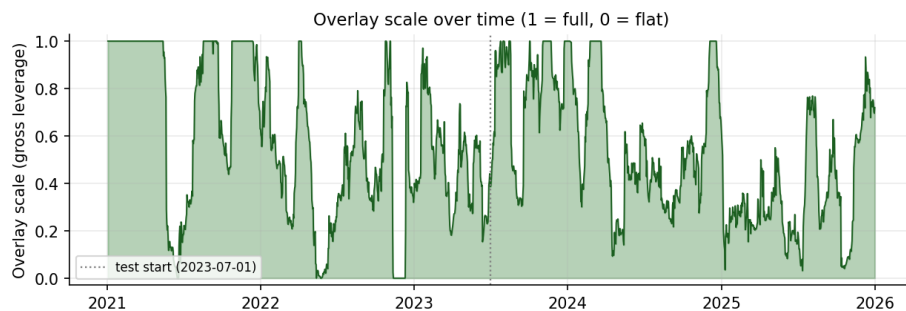


Figure 15. The resulting overlay scale (gross leverage) over time — full when correlation is low, flat when it spikes.

6. Where the Edge Comes From — Signal and Ensemble Alpha

To confirm that each signal earns its place through genuine *timing* (being positioned at the right moments) rather than mere market exposure, we hold each signal's positions fixed and shuffle the return dates 500 times to build a null distribution that destroys timing while preserving the return marginals. A signal whose true Sharpe sits in the top tail of that null carries real timing alpha.

Signal	Timing-alpha p-value	Role
Volume Momentum	0.002	alpha source
Vol-Adjusted Reversal	0.020	alpha source
Smart TSMOM	0.21	diversifier (long-side drift, not timing)

Volume Momentum and the reversal carry significant standalone timing alpha; Smart TSMOM does not, so it is kept for its diversification and long-drift capture rather than as a standalone alpha source.

Ensemble-level alpha. At the portfolio level the ensemble is essentially *market-neutral* — beta ≈ 0.01 and correlation ≈ 0.05 versus BTC — so its $\sim 16\%/yr$ return is **alpha** (market-neutral excess return), not disguised crypto-beta exposure. The block bootstrap puts the probability of a non-positive Sharpe at 2.5% (Section 7.2). Two of the three signals contribute genuine timing alpha, the third contributes diversification, and the combination delivers significant, near-zero-beta alpha.

7. Results & Robustness

7.1 Robustness of the reversal across overlay regimes

Sweeping the overlay aggressiveness, the 3-component ensemble beats the 2-component one in 6/6 configurations that deploy $\geq 30\%$ of capital (+0.06 to +0.29 Sharpe, with a better drawdown each time). The benefit grows with

deployment — an uncorrelated signal only diversifies when the ensemble is actually positioned — so the reversal's contribution is not an artifact of one overlay setting.

7.2 Stress tests

- **Walk-forward (3c ensemble):** 4/6 positive test sub-periods, mean SR 1.25 ± 1.80 .
- **vs naive baselines:** SR 1.56 vs BTC 1.15, BTC + vol-targeting 0.86, equal-weight 0.28; volatility ~11% vs BTC 46%, drawdown -11% vs -32%.
- **Transaction-cost stress:** SR 1.80 @ 10 bps, **1.56 @ 15 bps (headline)**, 1.31 @ 20 bps, 0.83 @ 30 bps; break-even $\approx$ 45 bps.
- **Bootstrap (1000 block-resamples, block = 20 days):** 95% CI [0.02, 2.99], $P(\text{SR} \leq 0) = 2.5\%$, $P(\text{SR} \leq 1) = 24\%$ — the edge is real but estimated with meaningful uncertainty.

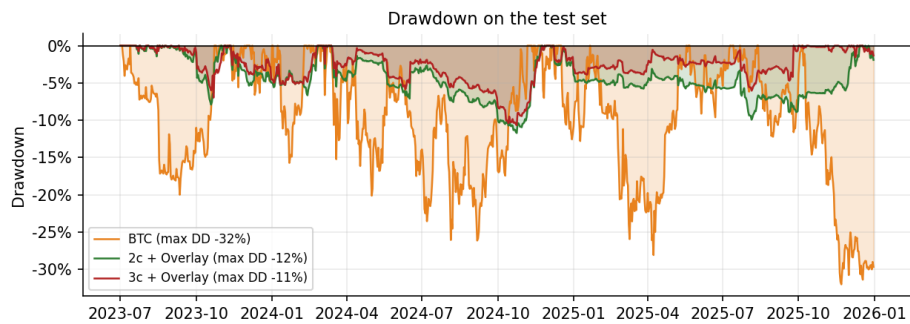


Figure 16. Drawdown on the test set — the 3c ensemble (dark red) is the shallowest.

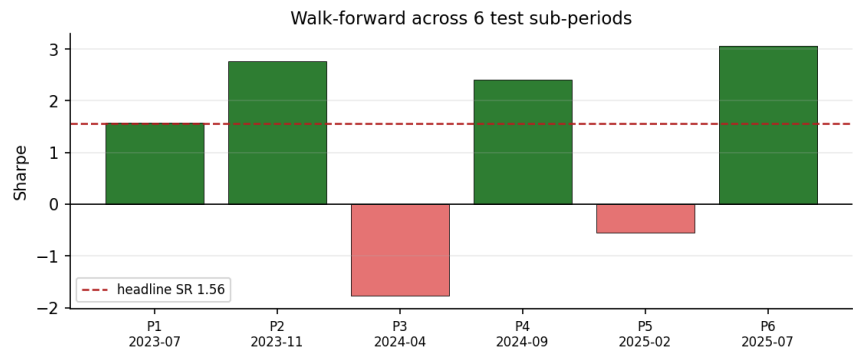


Figure 17. Sharpe across 6 disjoint test sub-periods (4 of 6 positive).

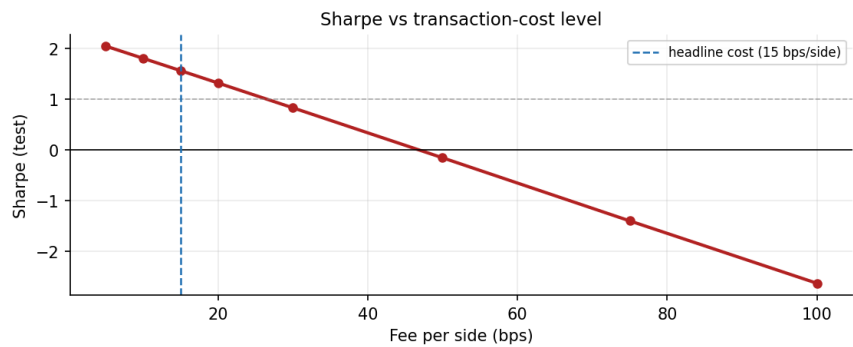


Figure 18. Sharpe vs transaction-cost level; break-even near 45 bps/side.

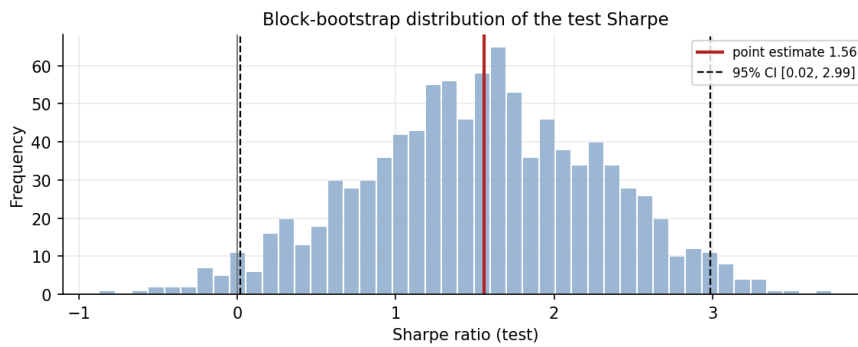


Figure 19. Block-bootstrap distribution of the test Sharpe (95% CI [0.02, 2.99]).

7.3 Risk-targeted return profile

The headline runs at ~11% volatility (~52% capital deployed). As Sharpe is leverage-invariant, the same return stream rescales to any risk target (real metrics of each compounded path shown):

Vol target	Leverage	Ann. return	CAGR	Max DD	Sharpe	Note
10.6%	1.0×	16.6%	17.4%	-10.7%	1.56	as presented, ~52% capital, no borrowing
15%	1.4×	23.4%	24.9%	-14.8%	1.56	~73% capital, no borrowing
20%	1.9×	31.1%	33.9%	-19.4%	1.56	~98% capital, no borrowing
30%	2.8×	46.7%	52.6%	-28.0%	1.56	needs leverage (funding cost)
46% (= BTC)	4.3×	71.6%	84.4%	-40.4%	1.56	matched to BTC vol; needs leverage

The 16.6% headline is the conservative low-vol point: simply deploying idle cash (no borrowing) reaches ~31%/yr at 20% vol. Levered rows are gross of perp-funding costs.

8. Negative Results

Each of the following was tested and discarded. The transferable lesson reinforces the design thesis: **out-of-sample robustness requires the signal's causal direction to be stable across regimes.**

Experiment	Out-of-sample result	Why it failed
Funding-rate binary (Schmeling et al. 2023)	SR test ≤ 0.2	Edge sign inverts between regimes
Stablecoin-supply macro signal	hurt the ensemble	Same regime-dependence
Universe expansion 53 $\rightarrow$ 76 (lower liquidity)	all metrics worse	Noise + slippage > diversification
Idiosyncratic momentum (Blitz et al. 2011)	SR test ≈ 0	Regime shift on residualized returns
Multi-horizon TSMOM basket (Hurst-Ooi-Pedersen 2017)	hurt the ensemble	Too correlated; no diversification
BTC-ETH / multi-pair stat-arb	SR test < 0	No mean reversion during a bull
Factor Zoo (continuous cross-sectional)	SR test ≈ -1.1	IC weights learned in bear invert in bull

Macro/sentiment/positioning signals are intrinsically regime-dependent (the market changes character between cycles); per-asset signals with mechanical causality (price $\rightarrow$ price, volume $\rightarrow$ price) survive.

9. Limitations

- **Single train/test split.** Walk-forward stability partially mitigates; an anchored, per-fold re-fit walk-forward would be the next level of validation.
- **Survivorship bias.** The universe requires data through the test period, excluding delisted coins — a real upward bias, not quantified (no delisting database).
- **The reversal is the lowest-confidence component** — a rare-event signal kept capped at 20%.
- **Wide Sharpe confidence interval** ([0.02, 2.99]); the lower bound is barely positive.
- **Research, not production:** no per-trade slippage modeling, capacity analysis, or funding costs in the levered profiles.

10. Conclusion

The strategy combines three economically distinct, near-orthogonal signals — trend (Smart TSMOM), volume conviction (Volume Momentum), and liquidity reversal — into an inverse-variance ensemble, then sizes it with a correlation-regime overlay. The result is a market-neutral, out-of-sample Sharpe of **1.56** net of realistic costs, beating BTC at roughly 4 $\times$ lower volatility. Its edge comes from two sources working together: signals with regime-stable causal mechanisms, and the diversification of combining low-correlation strategies. The return level is a risk-budget choice; the edge is the Sharpe.

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Appendix — Frozen Parameters

Every parameter is fixed on the train window (2021-01 → 2023-06) before a single test pass. What matters for overfitting is not the raw count but **how each value was chosen**: only parameters *optimized against train performance* consume degrees of freedom. The table tags each accordingly.

Component	Parameters	Type	How it was chosen
TSMOM	N=30	optimized	1-D train lookback scan (a sharp peak — disclosed)
TSMOM	threshold=0.02, vol_window=20	convention	standard TSMOM settings
Vol-managing	window=20, target_daily_vol=0.01	optimized	2-D train scan, top-1 of 64 cells
Smart TSMOM	sharpe_window=30, min_sharpe=0.3, long_alloc=0.70	literature	AdaptiveTrend (Nguyen 2026)
Volume Momentum	change_window=30, z_window=60, threshold=0.5, vol_window=20	convention	standard; sits on a robust plateau
Vol-Adj Reversal	lookback=4 (24h), k_σ=4.0	optimized	2-D train scan (rare-event, noisy surface)
Vol-Adj Reversal	vol_window=120/80	convention	standard realized-vol windows
IV ensemble	reversal cap 20%; inverse-variance weights	ex-ante / formula	cap set in advance; weights are a closed-form function of train variances, not tuned
Correlation overlay	(corr_low, corr_high) = (0.55, 0.85)	optimized (conservative)	1-standard-error rule on train — not the in-sample maximum

Effective degrees of freedom. The table lists ~18 numbers, but the count overstates the fitting freedom. Most values are set by **literature** (the three Smart-TSMOM filters) or **convention** (the Volume-Momentum and realized-vol windows), and the ensemble weights are a closed-form function of train variances — none of these is tuned to in-sample performance. Only **seven scalars** were actually optimized against the train Sharpe: TSMOM N; vol-managing window and target; reversal lookback and k_σ; and the overlay band corr_low, corr_high — and the last two were set by the conservative 1-standard-error rule rather than the train optimum. Seven fitted scalars against ~900 daily train observations (reinforced by a 53-coin cross-section per rule) is a very high data-to-parameter ratio.

Where selection bias lives — and why it does not drive the result. Picking the top cell of a 64-cell grid (vol-managing) and tuning a rare-event signal (reversal) genuinely inflate the in-sample Sharpe through multiple comparisons; we do not claim otherwise. Three things contain it. (i) Those are exactly the components that give the most back out-of-sample — vol-managed TSMOM decays from a train Sharpe of 1.62 to 0.72 on test — and they are **down-weighted**, while the primary alpha (Volume Momentum, ~52% of risk) sits on a parameter plateau rather than a tuned peak. (ii) The reversal is **capped at 20%** ex ante and flagged as the lowest-confidence component. (iii) Every layer is judged only on the untouched test set: the 3c ensemble carries a train Sharpe of **1.74** to a test Sharpe of **1.56** — a modest, expected give-back, not a collapse — and survives walk-forward (4 of 6 sub-periods positive) and a block-bootstrap whose 95% CI excludes zero. The edge therefore generalizes; the parameterization is not the source of a fragile result.

Full pipeline reproducible from `strategy.ipynb`.